



Sticks Kebab

Customer Segmentation & Location Recommendation

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Introduction



This project uses **kmeans clustering analysis** to uncover distinct **consumer segments** of Sticks Kebab. By analyzing each segment's core demographics, lifestyle traits, household structure and habits, we aim to identify **the most attractive target segments** and **the optimal new restaurant location** among the four proposed sites.

This analysis will support Sticks' expansion strategy by aligning its brand, strategic objectives, and revenue growth potentials with target consumers who value **healthy, convenient** and **family-friendly dining experiences**.

About Sticks Kebab

Overview



Est.: 2001

Format: Fast-casual

Cuisine: Mediterranean

Headquarter: Charlottesville, VA

Operating stores: 4

Core customer demographics



Families



Professionals



20-40s singles



Active



Healthy

Strategic objectives

- Build **loyalty** within the white-collar lunch crowd
- Convert loyal lunch customers to **frequent, dinner** ones
- Support **alternative growth** (retail hummus, off-site vending)

Restaurant concept



High quality

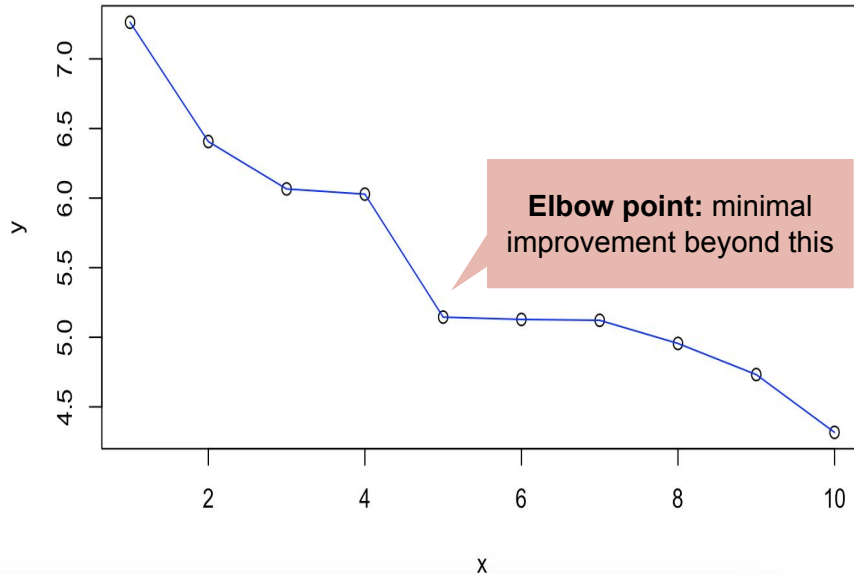
Healthy ingredients



Less stuffy enviroment

We began with the hierarchical clustering, using 5 base variables, namely D1.1, D1.2, D1.3, D1.3, D1.4 & Cust to identify the optimal number of segments i.e., 5

Elbow plot



- Based variables used:

Cust (1 = Customer, 0 = Non customer)

D1.1*: I tend to plan things very carefully

D1.2*: I sometimes have trouble controlling my spending

D1.3*: I think it is important to purchase products that are made locally

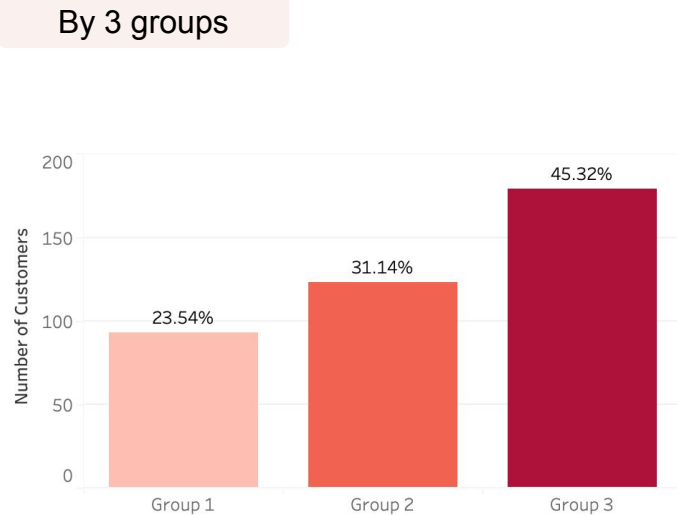
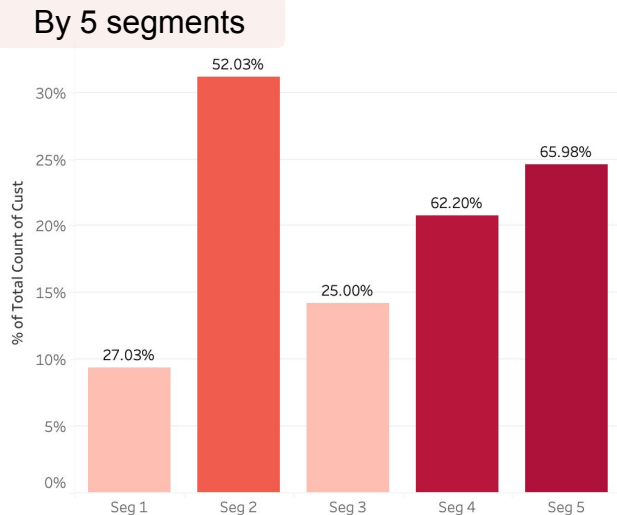
D1.4*: I carefully consider the health benefits of what I eat

- We choose these 5 variables as they capture key behavioral and attitudinal differences that distinguish customers from non-customers. Additionally, they do not contain any null values.
- 5 clusters were chosen at the elbow point where adding more clusters only yields a **small decrease** in within-cluster variance.
→ Diminishing returns in model improvement

*) For questions D1.1 to D1.4, respondent will answer with numerical responses from 1 to 5, with 1: Strongly Agree; 4: Strongly Disagree; 5: Don't Know

Using the k-means result, we mapped 395 respondents to their segments and consolidated these 5 into 3 main groups to better reflect their differences

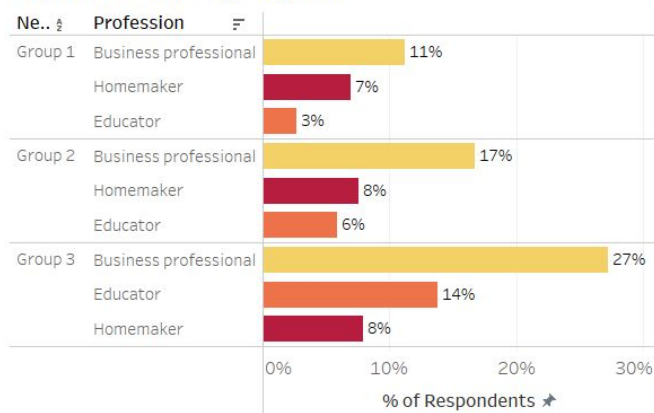
% Respondents to be customer in each segment



- Given similar customer likelihood across segments, we combined them into 3 main groups:
 - Group 1** (Segment 1: 27%, Segment 3: 25%) represents **smaller customer clusters** with relatively lower %
 - Group 2** (Segment 2: 52%) falls in between, representing a **moderate-sized customer group**
 - Group 3** (Segments 4: 62%, Segment 5: 66%) has the **largest customer bases**, signaling stronger market potential and higher engagement

Demographics: All groups are mainly business professionals, with Group 3 having the biggest families versus Groups 1 & 2's smaller households

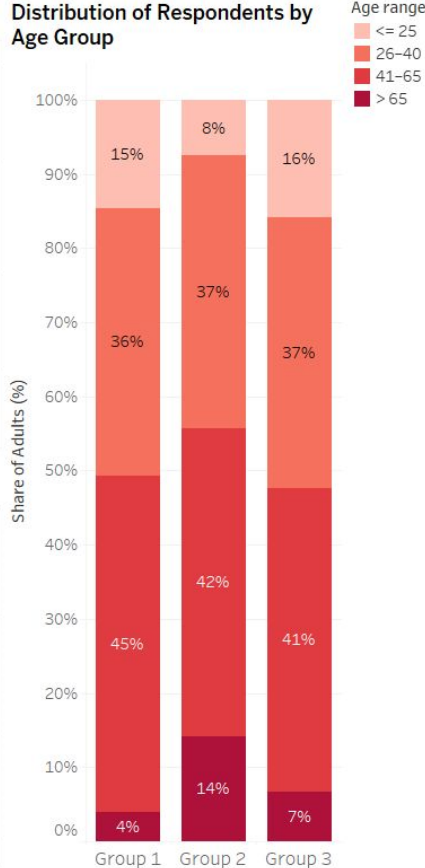
Top 3 Professions by Segment



Average number of children by age group



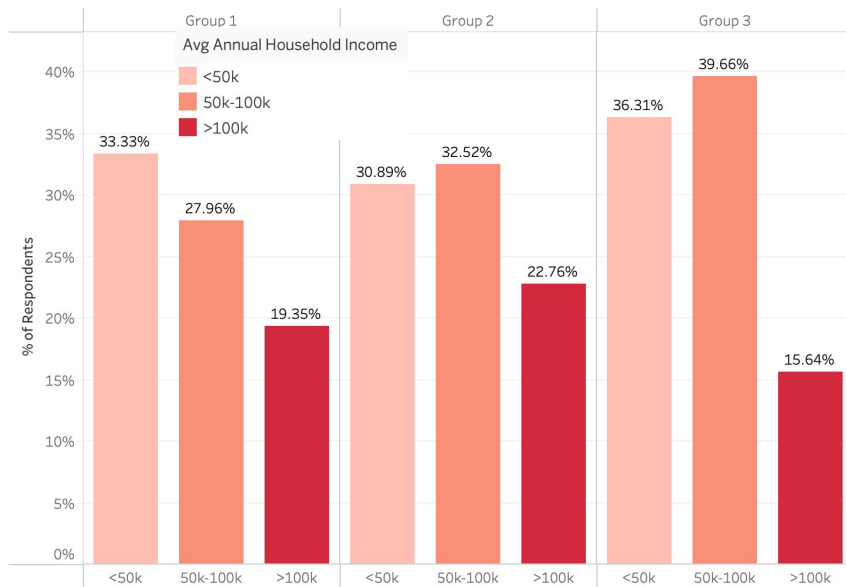
Distribution of Respondents by Age Group



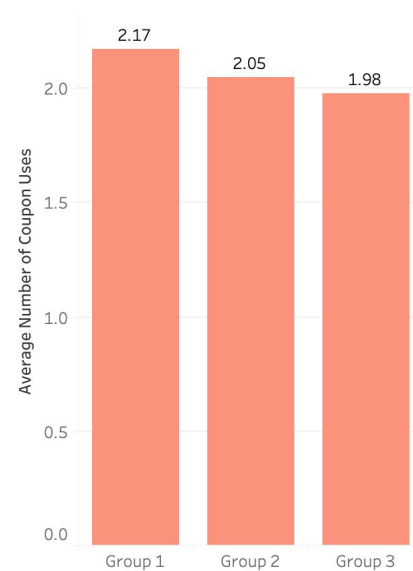
- **Group 3: Bigger household** with more children from all ages. Highest share from business professionals.
- **Groups 1 & 2: smaller or older** families with **fewer** professionals —> Steadier, more **traditional** lifestyles

Financial behaviors: While Group 1 is more price sensitive, Group 2 and 3 show more financial comfort with different levels of coupon usage

Average annual household income



Average coupon uses per month



- **Group 1:** Lower-income households (mostly <\$50k), highly price-sensitive & most reliant on coupons (~2.17 times/ month).
- **Group 2:** Higher-income families (mainly \$50–100k & large proportion of \$100k+), financially flexible with moderate coupon use (~2.05 times/ month).
- **Group 3:** Middle-income households (primarily \$50–100k), least reliant on coupons (avg 1.98 times a month).

Hobbies: Groups 2 & 3 are more socially, athletically, and family active, while Group 1 leads a more individualistic, less sport- or social event-oriented lifestyle

Respondents' hobbies

	Group 1	Group 2	Group 3
Eating out	47%	59%	54%
Volunteer work	25%	41%	32%
Attending local sporting events	22%	18%	30%
Kids' activities	18%	23%	22%
Running	17%	24%	21%
Hiking	18%	24%	21%
Yoga	13%	15%	18%
Bicycling	8%	15%	16%
Tennis	6%	5%	6%
Swimming	17%	9%	17%
Skiing/snowboarding	4%	3%	3%
Golf	5%	3%	4%
Gardening	20%	22%	23%
Fishing	10%	11%	13%
Camping	9%	12%	16%
Book clubs	6%	9%	11%
Visiting museums	23%	28%	30%
Other	23%	15%	19%

Their children's hobbies

	Group 1	Group 2	Group 3
Soccer	11%	15%	11%
Football	6%	7%	4%
Tennis	1%	7%	4%
Lacrosse	0%	7%	3%
Swimming	12%	17%	9%
Hiking	2%	15%	5%
Music	10%	16%	12%
Visiting local attractions	14%	28%	18%
Visiting children's museums	6%	16%	12%
Attending local sporting events	8%	24%	16%
Other	26%	41%	25%

Habits: Group 1 are lunch casual diners. Group 2 are social and leisure-oriented diners; Group 3 seek health-conscious, family-friendly meals between activities

Customer's average visits to SK by occasions

Times per month

	Group 1	Group 2	Group 3
Weekday lunch	4.1	2.9	3.4
Weekday dinner	3.6	3.5	3.1
Weekend lunch	3.9	4.0	4.0
Weekend dinner	2.8	4.3	4.1
Stick events	5.1	5.4	5.5
After school snacks or after sport practice/ event	5.7	5.5	5.9

- Group 1 visits most for weekday lunches → **Routine and convenience** driven
- Group 2 dines less frequently on weekdays but more on the weekend or events → **Social & occasions-based**
- Group 3 favors after-school or snack visits → **Family & community oriented**

Attitudinal scores by segment

1: Strongly Agree; 4: Strongly Disagree; 5: Don't Know

	Group 1	Group 2	Group 3
I tend to plan things very carefully	2.2	2.2	1.3
I sometimes have trouble controlling my spending	2.5	3.5	2.6
I think it is important to purchase products that are made locally	2.3	1.9	1.7
I carefully consider the health benefits of what I eat	2.7	1.6	1.5

- Group 1 is less concerned about health or food origin.
- Group 2 is **careful with spending**, while Group 3 prioritizes **health and locally sourced food**

Segment summary: We categorize the survey respondents into 3 main profiles: The budget-friendly, the wealthy socialist, and the family-bonding

**Group 1:
The budget friendly**

**Group 2:
The wealthy socialist**

**Group 3:
The family bonding**

% to be customer*	26%	52%	64%
Age	Middle-aged dominant; Lowest share of seniors	Oldest (92% age 25+, in which 53% age 40+)	Middle-aged dominant; but Highest share of young adults
Price sensitivity	High (heaviest coupon user)	Medium (mid coupon use)	Low (least coupon reliant)
# Children	Small families (Skew to teenagers)	New-nesters (skew to young kids)	Large families (skew to grown-ups)
Hobbies	More individualistic ; less sport, outdoors engagement	Most active in sports overall, eating out, volunteering	High participation in outdoor & community events
Dining habit at SK	Weekday lunch visitors → Routine, convenience based	Weekend/event diners → Social, occasion-driven	After-school/snack visitors → On-the-go oriented
Planning, spending habit	Less health or origin conscious	Spending-mindful; Health conscious	Careful planner; Health or origin conscious

*) Equate # Customers in each group / Total # respondent in each group

Target segments identification: Groups 2 & 3 stand out as their demographic, hobbies, and habits better align with SK’s strategic focus & core customer profile

	Group 1: The budget friendly	Group 2: The wealthy socialist	Group 3: The family bonding
Strategic goal alignment	✗ Weekday lunch focus not support weekend/event vist	✓ Can support weekend traffic & experiential events	✓ Can support repeat visits, food trailer, hummus launch
Brand image alignment	✗ Individualistic, less health- and family oriented	✓ Active and social lifestyle	✓ Highly family, outdoor, and health oriented
Revenue potentials	Limited Lower income, smaller & fewer meals	Medium-High Mid-high income & social nature can drive ticket size	High Big family & related habits drive repeat visits & tiket size
Target segments			

Location summary: Each site features one or a mix of distinct customer profiles: older affluent, tradition-focused, family-centered, and young tech-savvy adults

	Location A	Location B	Location C	Location D
Demo-graphic	• Mainly aged 45+ and 65+ • Highly educated white - collar, managers, retirees • Empty nesters, fewer young children	• Mainly aged 30-65; mix of working adults & retirees • Lower education levles • Many older couples or single-person homes	• 30-50 years old large families, mostly with kids • College-educated white-collar professionals and aging homeowners	• Mostly 25-40 year-old professionals • Young singles & couples, some with kids • College-educated • Highly tech savvy
Income	● High Median annual income: \$92,700	☾ Low Median annual income: \$31,900	● Middle Median annual income: \$55,700	● Upper-middle Median annual income: \$75,500
Lifestyle & values	• Family-oriented, empty-nester lifestyles • Value quality, wellness, & cultural engagement	• Tradition & community-focused, comfort driven • Price sensitive • Some brand-driven spender	• Family-driven, work-life balance, suburban lifestyle • Big spender on travel, dining, family experiences	• Urban socialites • Fast-paced, convenience-driven • Value connection & flexibility
Interests	• Upscale/ organic restos • Golf, travel, opera, concert • College sporting events with families • Community engagement	• Local fairs, church events, TV sports, neighborhood gatherings • Organic health food	• QSR goer • Local traveler • Sports online, outdoor activities, school events • Community engagement	• QSR goer, food-delivery • Some exposure to rivals e.g., Chipotle, Mellow Mushroom • Fitness, live music, media • Frequent travelers; early tech adopters

Target location: Location C's family-oriented residents who value community and are willing to spend for the experiences make them a stronger fit for Sticks Kebob

Evaluation

Location A

- **Aging** population 45 + and 65 + with fewer families
- **Too high-income** households → Prefer Upscale dining

→ **Less appeal** for SK's family-oriented, budget accessible meals

Location B

- **Low-income** → Budget dining inclined
- Working-class & **retiree**;
- Many single-person or older-couple homes mostly **without kids**
- **Traditional, family driven** habits

→ **Lower adoption** of new or premium offerings

✓ Location C

- Family-orientation → Fit SK's **wholesome, convenient offerings**
- Middle-income households → Value SK's **accessible meals**
- Active lifestyle & community engagement → Support **repeat visits & brand loyalty**
- Spending on dining & experiences → More **willing to try out** SK

Location D

- **Mostly singles & couple** → Not closely align with SK's family oriented brand values
- Affluent, tech savvy youngsters + urban **lifestyle** + existing' exposure to **competitors** → Likely **high trial, low brand stickiness**
- D's **convenience-driven** profile risks repeating SK's struggle to grow nights/ weekend traffic

Marketing approach: Sticks Kebobs can target Location C via a multi-channel campaign that highlights its core values to build awareness and grow its family base

Channel	Activity examples	Key messaging	Occasion
Digital	• “From farm to table” reel series for SK’s food production process • In-app loyalty program • Social campaigns for local community e.g., School break deal	• Healthy meals for family • Convenient (e.g., “Quick grab”) • Community-focused	• Weekdays → after-school snacks; • Weekends → family meals/events
Experiential	• Food trailers at events • In-store food sampling • Cooking demo at sports watching events e.g., Super Bowl streaming	• Fun, local, active lifestyle; family-friendly	• Weekends, local sports events, concerts, school/community events
Retail	• In-store promo • Free sample for repeat visit • Offer at vending machine off office building, school cafeteria	• Health-conscious, convenient, locally sourced	• Year-round, grocery shopping peaks
Traditional news	• News, magazine ads or features • Sponsor events e.g., kids’ sport league • Cultural festival presence • Paper coupon	• Community engagement; trusted family brand	• Seasonal/ local events